

ROZAS AP, Spain

WMO: 080080

Lat: 43.1114N

Lon: 7.4575W

Elev: 1463

StdP: 13.94

Time Zone: 1.00 (EUC)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.4	29.0	24.1	19.2	29.9	26.7	21.7	31.5	23.2	46.5	19.7	46.7	1.7	120	0.500

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	86.1	68.3	82.0	67.2	78.3	65.3	70.3	82.4	68.2	79.2	66.4	76.3	5.5	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.1	101.5	77.0	63.9	94.1	73.5	62.6	89.7	70.6	35.1	81.7	33.2	79.3	31.8	76.8	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	DB	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.9	16.1	13.3		WB	21.4	95.1	4.1	4.6	18.5	98.4	16.1	101.1	13.8	103.6	10.8	107.0
				20.8	75.1	4.4	2.2	17.7	76.7	15.1	78.0	12.7	79.2	9.5	80.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.6	42.8	44.5	46.3	50.9	54.1	61.0	65.7	65.1	62.7	56.0	48.8	45.1
	DBStd	9.67	5.85	5.04	4.50	5.32	6.06	5.59	4.43	4.97	5.20	4.97	6.29	5.72
	HDD50	885	230	164	133	51	28	2	0	0	0	9	98	170
	HDD65	4347	687	575	580	423	342	144	45	59	107	282	487	616
	CDD50	2208	8	9	18	78	154	332	486	469	380	194	61	19
	CDD65	198	0	0	0	0	4	25	66	63	37	3	0	0
	CDH74	1945	0	0	4	26	94	263	484	623	383	68	0	0
	CDH80	588	0	0	0	0	26	85	140	211	114	12	0	0
(14) Wind	WSAvg	4.8	4.9	5.1	6.2	6.0	4.8	4.7	4.6	4.4	4.1	3.7	4.1	4.7
(15) Precipitation	PrecAvg	41.2	5.2	3.7	3.2	4.1	3.1	1.7	1.2	1.4	2.6	5.3	5.4	4.7
	PrecMax	51.8	11.2	8.4	11.7	13.3	6.5	6.3	3.0	4.5	8.1	10.3	11.8	10.9
	PrecMin	26.7	1.1	0.7	0.0	0.7	0.7	0.0	0.0	0.3	0.4	0.5	1.1	0.7
	PrecStd	7.4	2.9	2.2	2.6	2.8	1.5	1.4	0.7	0.9	1.7	2.2	2.6	2.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.2	68.7	69.7	77.3	83.7	89.2	89.4	92.3	88.8	80.7	68.3	64.0
		MCWB	53.7	52.9	52.4	58.7	64.0	70.2	70.1	69.9	67.5	61.1	59.7	53.6
	2%	DB	55.9	60.3	62.9	72.2	75.0	81.4	83.5	85.1	82.9	74.8	63.8	61.0
		MCWB	52.2	50.5	52.8	58.1	63.0	67.9	69.2	68.4	65.6	61.4	57.9	52.3
	5%	DB	53.4	55.5	58.4	66.5	69.0	75.6	79.3	80.7	78.0	69.9	61.0	57.5
		MCWB	49.9	50.1	50.4	55.6	59.2	65.2	67.4	66.9	64.8	60.1	56.5	52.0
	10%	DB	51.3	53.0	54.3	61.1	64.5	71.2	75.2	76.0	73.4	65.7	58.1	54.5
		MCWB	48.2	49.1	48.7	52.9	56.8	62.7	65.8	65.1	63.4	58.6	54.6	51.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.8	54.7	56.1	60.8	66.9	73.7	72.7	73.3	69.4	65.1	62.0	56.3
		MCDB	57.2	61.9	66.5	73.7	79.0	87.2	84.5	84.8	82.5	72.9	66.0	60.1
	2%	WB	52.9	53.0	53.4	58.5	63.5	68.6	70.0	69.6	67.2	62.6	59.5	54.0
		MCDB	54.7	57.0	60.4	70.4	74.2	77.8	81.7	81.9	79.4	70.1	63.2	57.1
	5%	WB	50.7	51.3	51.6	56.2	60.4	65.6	67.9	67.4	65.4	61.1	57.0	52.8
		MCDB	52.7	55.5	57.2	65.3	67.7	74.9	78.0	78.5	76.0	68.1	60.3	56.2
	10%	WB	48.6	49.3	49.7	53.7	57.4	63.3	65.9	65.5	63.4	59.2	54.8	51.3
		MCDB	50.8	52.9	54.2	60.3	63.1	70.4	74.3	74.8	71.8	65.9	57.4	55.0
(35) Mean Daily Temperature Range	5% DB	MDBR	12.5	14.7	16.3	18.2	17.5	17.3	18.9	21.9	21.9	19.7	13.3	13.4
		MCDBR	14.0	21.9	25.1	29.7	27.1	27.4	27.0	30.6	30.3	28.3	16.9	18.1
		MCWBR	10.5	14.0	15.2	16.7	15.4	14.7	14.0	14.9	13.9	16.4	11.6	12.3
	5% WB	MCDBR	10.4	19.0	22.0	27.2	26.3	24.3	24.8	28.2	27.0	24.5	14.5	14.3
		MCWBR	8.6	13.1	14.0	15.9	15.6	13.8	13.5	14.5	13.5	15.0	10.7	10.9
(40) Clear-Sky Solar Irradiance	taub		0.315	0.325	0.358	0.369	0.376	0.381	0.368	0.368	0.360	0.348	0.324	0.309
	taud		2.502	2.456	2.365	2.340	2.344	2.349	2.385	2.394	2.417	2.465	2.497	2.512
	Ebn at Noon		261	277	279	285	285	282	285	281	275	265	255	252
	Edh at Noon		23	28	34	38	39	39	37	36	32	27	23	21
(44) All-Sky Solar Radiation	RadAvg		461	737	1095	1446	1739	1939	1962	1782	1400	885	530	422
(45)	RadStd		61	109	167	172	168	148	121	120	76	107	83	60

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page